Processing constraints and resource competition in non-native speech error detection

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Listeners often rely on heuristics rather than fully verifying linguistic input, as evidenced in classic demonstrations like the *Moses Illusion,* where semantic anomalies go unnoticed [1-3]. Prior work shows such failures are exacerbated when cognitive resources are constrained [4]. Critically, recent reviews (e.g., [5]) suggest a tension between task and resource capacity: even when a task explicitly demands deep verification, excessive cognitive load can force a fallback to underspecified, heuristic processing. Non-native speech presents a natural testbed for this "load-induced fallback." Processing non-native pronunciation imposes a high cognitive demand (e.g., [6]). At the same time, naïve, native listeners often fail to detect higher-level linguistic violations in non-native speech [7, 8].

Pursued in various lines of work, these findings leave open a fundamental question: **how do listeners allocate attention resources when the signal contains *multiple* potential sources of deviation?** Does the processing cost of a non-native accent trigger a global reduction in error detection due to resource overload [1], or does it invite a selective, adaptive response based on listeners' inference of the reliability of each source of information? [2]. To test this, we systematically manipulated syntactic/lexical violations (e.g., *doing→making homework*) and phonological violations (e.g., *very→berry much*), using naturalistic examples from non-native speech. We ask how both error types—alone or in combination—*consume resources and affect error detection* (using task from [4]). Specifically, we test two related questions:

Q1: Detectability of different error types? It is possible that bottom-up processing of accented speech takes precedence [9, 10], limiting resources for subsequent higher-level processing, which dampens detection of syntactic/lexical errors (Hypothesis 1a). However, *non-native* speech may trigger greater reliance on top-down predictions, weakening bottom-up encoding [12] and leading to under-detection of phonological errors (Hypothesis 1b).

Q2: Resource Allocation: Prioritization vs. Competition? Listeners may **prioritize** detecting one error type, regardless of the presence or absence of the other error type (Hyp2a). Alternatively, listeners may experience resource **competition** (Hyp2b), where co-occurring error types drain a shared resource pool, reducing detection rate for both error types.

Experiment: U.S.-born L1 English speakers (n=46) listened to trivia questions (*see ex.1*) in Spanish-accented English. Participants heard each question twice and had two attempts to report errors or answer the question if error-free. We manipulated whether questions contained (i) no error, (ii) a phonological error, (iii) a non-phonological (syntactic/lexical) error or (iv) both error types. Trial-level accuracy was analyzed via GLMER, fitted separately for Attempt 1 (initial processing) and Combined attempts (after re-exposure) (*see 'Model specifications'*).

Results show clear differences between Attempts 1 and 2. **Q1 Detectability.** While there was no difference between error types (*d'*) on Attempt 1, combined-attempts results support **Hyp1b.** Upon re-exposure, listeners detected more non-phonological errors compared to phonological ones ($\mathrm{ErrorType} \times \mathrm{Present}$, *p<.05, Fig.1*). This suggests non-phonological errors are easier to detect once the initial processing load is alleviated. **Q2 Resource allocation.** On Attempt 1, results support **Hyp2b**: When both error types were present, detection of either error type was lower than when they appeared alone (main effect of $\mathrm{OtherTypePresent}$; *p<.05, Fig.2*). Crucially, this masking effect applies equally to both error types ($\mathrm{ErrorType} \times \mathrm{OtherTypePresent}$; *p=n.s.*), suggesting that processing multiple deviations drains shared resources during initial processing. No effects were significant in combined attempts.

In sum, these findings suggest non-native speech error detection is constrained by resource capacity limits. Initial processing is characterized by **competition**, where coexisting error types mask one another. Once resource constraints are alleviated by **re-exposure,** listeners demonstrate greater sensitivity to higher-level linguistic deviations than phonological deviations.

Model specifications

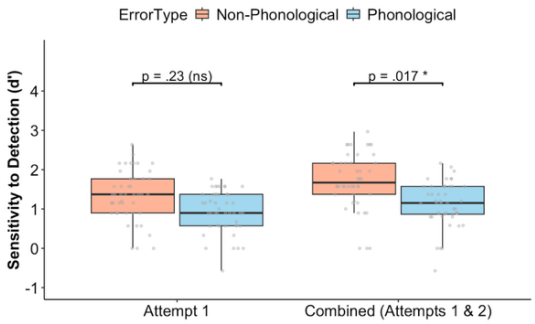
$$\text{Reported} \sim \text{ErrorType} \times \text{Present} + \text{ ErrorType} \times \text{OtherTypePresent} + (1|\text{Participant}) + (1|\text{Item})$$

Model includes three fixed effects and two interactions, every trial coded for presence of both error types simultaneously. Specified Reported as the dependent variable, tracking whether the participant successfully identified the specific error type being evaluated. Fixed effects were ErrorType (sum-coded: 1 = Non-Phonological, -1 = Phonological), Present (if the target error was actually in the stimulus), and OtherTypePresent (if the other error type was co-occurring).

ex. 1. Experimental design *(topics varied; types of phonological and non-phonological errors are from prior work and are naturally occurring in Spanish-accented English; all minimal pairs differed in part-of-speech; error locations counterbalanced; 24 targets, 32 fillers, Latin-Square; recorded by a female L1 Peninsula-Spanish speaker; instructions and practice items explain questions could involve multiple errors; question presented auditorily).*

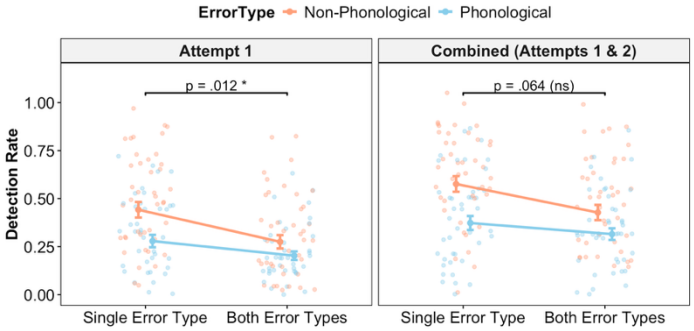
No error: For what pink ice cream flavor will companies do their best to use real fruit?
Phon error: For what pink ice cream flavor **wheel** companies do their best to use real fruit?
Non-Phon error: For what pink ice cream flavor will companies *make* their best to use real fruit?
Both errors: For what pink ice cream flavor **wheel** companies *make* their best to use real fruit?

Fig. 1. Detectability of Phonological and Non-Phonological Errors



Note. Comparison of sensitivity between Attempt 1 and the Combined Attempts. The interaction between Error Type and Attempt reveals that while detection rates were similar initially, listeners were significantly better at detecting non-phonological errors upon re-exposure.

Fig. 2. Does detection rate decrease when a second error type is present?



Note. The downward slope indicates the "cost" of adding a second error (masking). Results show that masking was significant during the first attempt but was alleviated in the combined attempts, suggesting a release from resource competition.

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